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Blood control for a catheter insertion device

Abstract

A fluid control component configured for controlling fluid flow through the hub of a catheter assembly during and after placement into the patient is disclosed. In one embodiment, the fluid control component comprises a body disposed within a cavity of the hub, the body being movable between a first position and a second position, wherein the body does not pierce a valve disposed in the hub when in the first position and wherein the body pierces the valve when in the second position. The body includes a conduit to enable fluid flow through an internal portion of the body when the body is in the second position, and a plurality of longitudinal ribs disposed on an exterior surface of the body. The longitudinal ribs can provide fluid flow channels between the valve and an external portion of the body when the body is in the second position.

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5984895	12/1998	Padilla et al.	N/A	N/A
5984903	12/1998	Nadal	N/A	N/A
5989220	12/1998	Shaw et al.	N/A	N/A
5989271	12/1998	Bonnette et al.	N/A	N/A
5997507	12/1998	Dysarz	N/A	N/A
5997510	12/1998	Schwemberger	N/A	N/A
6001080	12/1998	Kuracina et al.	N/A	N/A
6004278	12/1998	Botich et al.	N/A	N/A
6004294	12/1998	Brimhall et al.	N/A	N/A
6004295	12/1998	Langer et al.	N/A	N/A
6011988	12/1999	Lynch et al.	N/A	N/A
6019736	12/1999	Avellanet et al.	N/A	N/A
6022319	12/1999	Willard et al.	N/A	N/A
6024727	12/1999	Thome et al.	N/A	N/A
6024729	12/1999	Dehdashtian	604/167.04	A61M 39/0606
6045734	12/1999	Luther et al.	N/A	N/A
6056726	12/1999	Isaacson	N/A	N/A
6059484	12/1999	Greive	N/A	N/A
6066100	12/1999	Willard et al.	N/A	N/A
6074378	12/1999	Mouri et al.	N/A	N/A
6077244	12/1999	Botich et al.	N/A	N/A
6080137	12/1999	Pike	N/A	N/A
6083237	12/1999	Huitema et al.	N/A	N/A
6096004	12/1999	Meglan et al.	N/A	N/A
6096005	12/1999	Botich et al.	N/A	N/A
6109264	12/1999	Sauer	N/A	N/A
6117108	12/1999	Woehr et al.	N/A	N/A
6120494	12/1999	Jonkman	N/A	N/A
6126633	12/1999	Kaji et al.	N/A	N/A
6126641	12/1999	Shields	N/A	N/A
6139532	12/1999	Howell et al.	N/A	N/A
6139557	12/1999	Passafaro et al.	N/A	N/A
6159179	12/1999	Simonson	N/A	N/A

6171234	12/2000	White et al.	N/A	N/A
6171287	12/2000	Lynn et al.	N/A	N/A
6176842	12/2000	Tachibana et al.	N/A	N/A
6193690	12/2000	Dysarz	N/A	N/A
6197001	12/2000	Wilson et al.	N/A	N/A
6197007	12/2000	Thome et al.	N/A	N/A
6197041	12/2000	Shichman et al.	N/A	N/A
6203527	12/2000	Zadini et al.	N/A	N/A
6213978	12/2000	Voyten	N/A	N/A
6217558	12/2000	Zadini et al.	N/A	N/A
6221047	12/2000	Greene et al.	N/A	N/A
6221048	12/2000	Phelps	N/A	N/A
6221049	12/2000	Selmon et al.	N/A	N/A
6224569	12/2000	Brimhall	N/A	N/A
6228060	12/2000	Howell	N/A	N/A
6228062	12/2000	Howell et al.	N/A	N/A
6228073	12/2000	Noone et al.	N/A	N/A
6245045	12/2000	Stratienko	N/A	N/A
6251092	12/2000	Qin et al.	N/A	N/A
6268399	12/2000	Hultine et al.	N/A	N/A
6270480	12/2000	Dorr et al.	N/A	N/A
6273871	12/2000	Davis et al.	N/A	N/A
6280419	12/2000	Vojtasek	N/A	N/A
6287278	12/2000	Woehr et al.	N/A	N/A
6309379	12/2000	Willard et al.	N/A	N/A
6319244	12/2000	Suresh et al.	N/A	N/A
6322537	12/2000	Chang	N/A	N/A
D452003	12/2000	Niermann	N/A	N/A
6325781	12/2000	Takagi et al.	N/A	N/A
6325797	12/2000	Stewart et al.	N/A	N/A
6336914	12/2001	Gillespie, III	N/A	N/A
6352520	12/2001	Miyazaki	N/A	N/A
6368337	12/2001	Kieturakis et al.	N/A	N/A
6379333	12/2001	Brimhall et al.	N/A	N/A
6379372	12/2001	Dehdashtian et al.	N/A	N/A
D457955	12/2001	Bilitz	N/A	N/A
6406442	12/2001	McFann et al.	N/A	N/A
D460179	12/2001	Isoda et al.	N/A	N/A
6422989	12/2001	Hektner	N/A	N/A
6436070	12/2001	Botich et al.	N/A	N/A
6436112	12/2001	Wensel et al.	N/A	N/A
6443929	12/2001	Kuracina et al.	N/A	N/A
6451052	12/2001	Burmeister et al.	N/A	N/A
6461362	12/2001	Halseth et al.	N/A	N/A
6475217	12/2001	Platt	N/A	N/A
6478779	12/2001	Hu	N/A	N/A
6485473	12/2001	Lynn	N/A	N/A
6485497	12/2001	Wensel et al.	N/A	N/A
6497681	12/2001	Brenner	N/A	N/A
6506181	12/2002	Meng et al.	N/A	N/A

6514236	12/2002	Stratienko	N/A	N/A
6524276	12/2002	Halseth et al.	N/A	N/A
D471980	12/2002	Caizza	N/A	N/A
6527759	12/2002	Tachibana et al.	N/A	N/A
6530913	12/2002	Giba et al.	N/A	N/A
6530935	12/2002	Wensel et al.	N/A	N/A
6540725	12/2002	Ponzi	N/A	N/A
6540732	12/2002	Botich et al.	N/A	N/A
6544239	12/2002	Kinsey et al.	N/A	N/A
6547762	12/2002	Botich	604/110	A61M 25/09041
6558355	12/2002	Metzger et al.	N/A	N/A
6582402	12/2002	Erskine	N/A	N/A
6582440	12/2002	Brumbach	N/A	N/A
6585703	12/2002	Kassel et al.	N/A	N/A
6589262	12/2002	Honebrink et al.	N/A	N/A
6595955	12/2002	Ferguson et al.	N/A	N/A
6595959	12/2002	Stratienko	N/A	N/A
6599296	12/2002	Gillick et al.	N/A	N/A
6607511	12/2002	Halseth et al.	N/A	N/A
6616630	12/2002	Woehr et al.	N/A	N/A
6620136	12/2002	Pressly, Sr. et al.	N/A	N/A
6623449	12/2002	Paskar	N/A	N/A
6623456	12/2002	Holdaway et al.	N/A	N/A
6626868	12/2002	Prestidge et al.	N/A	N/A
6626869	12/2002	Bint	N/A	N/A
6629959	12/2002	Kuracina et al.	N/A	N/A
6632201	12/2002	Mathias et al.	N/A	N/A
6638252	12/2002	Moulton et al.	N/A	N/A
6641564	12/2002	Kraus	N/A	N/A
6645178	12/2002	Junker et al.	N/A	N/A
6652486	12/2002	Bialecki et al.	N/A	N/A
6652490	12/2002	Howell	N/A	N/A
6663577	12/2002	Jen et al.	N/A	N/A
6663592	12/2002	Rhad et al.	N/A	N/A
6666865	12/2002	Platt	N/A	N/A
6679900	12/2003	Kieturakis et al.	N/A	N/A
6689102	12/2003	Greene	N/A	N/A
6692508	12/2003	Wensel et al.	N/A	N/A
6692509	12/2003	Wensel et al.	N/A	N/A
6695814	12/2003	Greene et al.	N/A	N/A
6695856	12/2003	Kieturakis et al.	N/A	N/A
6695860	12/2003	Ward et al.	N/A	N/A
6699221	12/2003	Vaillancourt	N/A	N/A
6702811	12/2003	Stewart et al.	N/A	N/A
6706018	12/2003	Westlund et al.	N/A	N/A
6711444	12/2003	Koblish	N/A	N/A
6712790	12/2003	Prestidge et al.	N/A	N/A
6712797	12/2003	Southern, Jr.	N/A	N/A
6716197	12/2003	Svendsen	N/A	N/A

6730062	12/2003	Hoffman et al.	N/A	N/A
6740063	12/2003	Lynn	N/A	N/A
6740096	12/2003	Teague et al.	N/A	N/A
6745080	12/2003	Koblish	N/A	N/A
6749588	12/2003	Howell et al.	N/A	N/A
6764468	12/2003	East	N/A	N/A
D494270	12/2003	Reschke	N/A	N/A
6776788	12/2003	Klint et al.	N/A	N/A
6796962	12/2003	Ferguson et al.	N/A	N/A
6824545	12/2003	Sepetka et al.	N/A	N/A
6832715	12/2003	Eungard et al.	N/A	N/A
6835190	12/2003	Nguyen	N/A	N/A
6837867	12/2004	Kortelling	N/A	N/A
6860871	12/2004	Kuracina et al.	N/A	N/A
6872193	12/2004	Shaw et al.	N/A	N/A
6887220	12/2004	Hogendijk	N/A	N/A
6902546	12/2004	Ferguson	N/A	N/A
6905483	12/2004	Newby et al.	N/A	N/A
6913595	12/2004	Mastorakis	N/A	N/A
6916311	12/2004	Vojtasek	N/A	N/A
6921386	12/2004	Shue et al.	N/A	N/A
6921391	12/2004	Barker et al.	N/A	N/A
6929624	12/2004	Del Castillo	N/A	N/A
6939325	12/2004	Haining	N/A	N/A
6942652	12/2004	Pressly, Sr. et al.	N/A	N/A
6953448	12/2004	Moulton et al.	N/A	N/A
6958054	12/2004	Fitzgerald	N/A	N/A
6958055	12/2004	Donnan et al.	N/A	N/A
6960191	12/2004	Howlett et al.	N/A	N/A
6972002	12/2004	Thorne	N/A	N/A
6974438	12/2004	Shekalim	N/A	N/A
6994693	12/2005	Tal	N/A	N/A
7001396	12/2005	Glazier et al.	N/A	N/A
7004927	12/2005	Ferguson et al.	N/A	N/A
7008404	12/2005	Nakajima	N/A	N/A
7018372	12/2005	Casey et al.	N/A	N/A
7018390	12/2005	Turovskiy et al.	N/A	N/A
7025746	12/2005	Tal	N/A	N/A
7029467	12/2005	Currier et al.	N/A	N/A
7033335	12/2005	Haarala et al.	N/A	N/A
7044935	12/2005	Shue et al.	N/A	N/A
7060055	12/2005	Wilkinson et al.	N/A	N/A
7090656	12/2005	Botich et al.	N/A	N/A
7094243	12/2005	Mulholland et al.	N/A	N/A
7097633	12/2005	Botich et al.	N/A	N/A
7125396	12/2005	Leinsing et al.	N/A	N/A
7125397	12/2005	Woehr et al.	N/A	N/A
7141040	12/2005	Lichtenberg	N/A	N/A
7153276	12/2005	Barker et al.	N/A	N/A
7163520	12/2006	Bernard et al.	N/A	N/A

7169159	12/2006	Green et al.	N/A	N/A
7179244	12/2006	Smith et al.	N/A	N/A
7186239	12/2006	Woehr	N/A	N/A
7191900	12/2006	Opie et al.	N/A	N/A
7192433	12/2006	Osypka et al.	N/A	N/A
7204813	12/2006	Shue et al.	N/A	N/A
7214211	12/2006	Woehr et al.	N/A	N/A
7255685	12/2006	Pressly, Sr. et al.	N/A	N/A
7264613	12/2006	Woehr et al.	N/A	N/A
7291130	12/2006	McGurk	N/A	N/A
7303547	12/2006	Pressly, Sr. et al.	N/A	N/A
7303548	12/2006	Rhad et al.	N/A	N/A
7314462	12/2007	O'Reagan et al.	N/A	N/A
7331966	12/2007	Soma et al.	N/A	N/A
7344516	12/2007	Erskine	N/A	N/A
7354422	12/2007	Riesenberger et al.	N/A	N/A
7374554	12/2007	Menzi et al.	N/A	N/A
7381205	12/2007	Thommen	N/A	N/A
7396346	12/2007	Nakajima	N/A	N/A
7413562	12/2007	Ferguson et al.	N/A	N/A
7422572	12/2007	Popov et al.	N/A	N/A
7458954	12/2007	Ferguson et al.	N/A	N/A
7465294	12/2007	Vladimirsky	N/A	N/A
7468057	12/2007	Ponzi	N/A	N/A
7470254	12/2007	Basta	604/167.04	A61M 25/09041
7491176	12/2008	Mann	N/A	N/A
7494010	12/2008	Opie et al.	N/A	N/A
7500965	12/2008	Menzi et al.	N/A	N/A
7507222	12/2008	Cindrich et al.	N/A	N/A
7513887	12/2008	Halseth et al.	N/A	N/A
7513888	12/2008	Sircom et al.	N/A	N/A
7524306	12/2008	Botich et al.	N/A	N/A
7530965	12/2008	Villa et al.	N/A	N/A
7534227	12/2008	Kulli	N/A	N/A
7534231	12/2008	Kuracina et al.	N/A	N/A
7544170	12/2008	Williams et al.	N/A	N/A
7556617	12/2008	Voorhees, Jr. et al.	N/A	N/A
7566323	12/2008	Chang	N/A	N/A
D601243	12/2008	Bierman et al.	N/A	N/A
7597681	12/2008	Sutton et al.	N/A	N/A
7608057	12/2008	Woehr et al.	N/A	N/A
D604839	12/2008	Crawford et al.	N/A	N/A
7611485	12/2008	Ferguson	N/A	N/A
7611487	12/2008	Woehr et al.	N/A	N/A
7611499	12/2008	Woehr et al.	N/A	N/A
7618395	12/2008	Ferguson	N/A	N/A
7625360	12/2008	Woehr et al.	N/A	N/A
7628769	12/2008	Grandt et al.	N/A	N/A
7632243	12/2008	Bialecki et al.	N/A	N/A

7645263	12/2009	Angel et al.	N/A	N/A
7654988	12/2009	Moulton et al.	N/A	N/A
7658725	12/2009	Bialecki et al.	N/A	N/A
D612043	12/2009	Young et al.	N/A	N/A
7678080	12/2009	Shue et al.	N/A	N/A
7682358	12/2009	Gullickson et al.	N/A	N/A
7691088	12/2009	Howell	N/A	N/A
7691090	12/2009	Belley et al.	N/A	N/A
7691093	12/2009	Brimhall	N/A	N/A
7695458	12/2009	Belley et al.	N/A	N/A
7699807	12/2009	Faust et al.	N/A	N/A
D615197	12/2009	Koh et al.	N/A	N/A
7708721	12/2009	Khaw	N/A	N/A
7713243	12/2009	Hillman	N/A	N/A
7717875	12/2009	Knudson et al.	N/A	N/A
7722567	12/2009	Tal	N/A	N/A
7722569	12/2009	Soderholm et al.	N/A	N/A
D617893	12/2009	Bierman et al.	N/A	N/A
7731687	12/2009	Menzi et al.	N/A	N/A
7731691	12/2009	Cote et al.	N/A	N/A
7736332	12/2009	Carlyon et al.	N/A	N/A
7736337	12/2009	Diep et al.	N/A	N/A
7736339	12/2009	Woehr et al.	N/A	N/A
7736342	12/2009	Abriles et al.	N/A	N/A
7740615	12/2009	Shaw et al.	N/A	N/A
7744574	12/2009	Pederson et al.	N/A	N/A
7753877	12/2009	Bialecki et al.	N/A	N/A
7753887	12/2009	Botich et al.	N/A	N/A
7762984	12/2009	Kumoyama et al.	N/A	N/A
7762993	12/2009	Perez	N/A	N/A
7766879	12/2009	Tan et al.	N/A	N/A
7776052	12/2009	Greenberg et al.	N/A	N/A
7785296	12/2009	Muskatello et al.	N/A	N/A
7794424	12/2009	Paskar	N/A	N/A
7798994	12/2009	Brimhall	N/A	N/A
7803142	12/2009	Longson et al.	N/A	N/A
7828773	12/2009	Swisher et al.	N/A	N/A
7828774	12/2009	Harding et al.	N/A	N/A
7833201	12/2009	Carlyon et al.	N/A	N/A
7850644	12/2009	Gonzalez et al.	N/A	N/A
7857770	12/2009	Raulerson et al.	N/A	N/A
D634843	12/2010	Kim et al.	N/A	N/A
7896862	12/2010	Long et al.	N/A	N/A
7905857	12/2010	Swisher	N/A	N/A
7914488	12/2010	Dickerson	N/A	N/A
7914492	12/2010	Heuser	N/A	N/A
7922696	12/2010	Tal et al.	N/A	N/A
7922698	12/2010	Riesenberger et al.	N/A	N/A
7927314	12/2010	Kuracina et al.	N/A	N/A
7935080	12/2010	Howell et al.	N/A	N/A

7959613	12/2010	Rhad et al.	N/A	N/A
7972313	12/2010	Woehr et al.	N/A	N/A
7972324	12/2010	Quint	N/A	N/A
D643531	12/2010	van der Weiden	N/A	N/A
8029470	12/2010	Whiting et al.	N/A	N/A
8029472	12/2010	Leinsing et al.	N/A	N/A
8048031	12/2010	Shaw et al.	N/A	N/A
8048039	12/2010	Carlyon et al.	N/A	N/A
8057404	12/2010	Fujiwara et al.	N/A	N/A
8062261	12/2010	Adams	N/A	N/A
8075529	12/2010	Nakajima et al.	N/A	N/A
8079979	12/2010	Moorehead	N/A	N/A
D653329	12/2011	Lee-Sepsick	N/A	N/A
8100858	12/2011	Woehr et al.	N/A	N/A
8105286	12/2011	Anderson et al.	N/A	N/A
8105315	12/2011	Johnson et al.	N/A	N/A
8123727	12/2011	Luther et al.	N/A	N/A
8152758	12/2011	Chan et al.	N/A	N/A
8162881	12/2011	Lilley, Jr. et al.	N/A	N/A
8167851	12/2011	Sen	N/A	N/A
8177753	12/2011	Vitullo et al.	N/A	N/A
RE43473	12/2011	Newby et al.	N/A	N/A
8192402	12/2011	Anderson et al.	N/A	N/A
8202251	12/2011	Bierman et al.	N/A	N/A
8202253	12/2011	Wexler	N/A	N/A
8206343	12/2011	Racz	N/A	N/A
8211070	12/2011	Woehr et al.	N/A	N/A
8221387	12/2011	Shelso et al.	N/A	N/A
8226612	12/2011	Nakajima	N/A	N/A
8235945	12/2011	Baid	N/A	N/A
8251923	12/2011	Carrez et al.	N/A	N/A
8251950	12/2011	Albert et al.	N/A	N/A
D667111	12/2011	Robinson	N/A	N/A
8257322	12/2011	Koehler et al.	N/A	N/A
8273054	12/2011	St. Germain et al.	N/A	N/A
8286657	12/2011	Belley et al.	N/A	N/A
8298186	12/2011	Popov	N/A	N/A
8303543	12/2011	Abulhaj	N/A	N/A
8308685	12/2011	Botich et al.	N/A	N/A
8308691	12/2011	Woehr et al.	N/A	N/A
D672456	12/2011	Lee-Sepsick	N/A	N/A
8323249	12/2011	White et al.	N/A	N/A
8328762	12/2011	Woehr et al.	N/A	N/A
8328837	12/2011	Binmoeller	N/A	N/A
8333735	12/2011	Woehr et al.	N/A	N/A
8337424	12/2011	Palmer et al.	N/A	N/A
8337463	12/2011	Woehr et al.	N/A	N/A
8337471	12/2011	Baid	N/A	N/A
D675318	12/2012	Luk et al.	N/A	N/A
8361020	12/2012	Stout	N/A	N/A

8361038	12/2012	McKinnon et al.	N/A	N/A
8376994	12/2012	Woehr et al.	N/A	N/A
8377006	12/2012	Tal et al.	N/A	N/A
8382721	12/2012	Woehr et al.	N/A	N/A
8388583	12/2012	Stout et al.	N/A	N/A
8403886	12/2012	Bialecki et al.	N/A	N/A
8412300	12/2012	Sonderegger	N/A	N/A
8414539	12/2012	Kuracina et al.	N/A	N/A
8419688	12/2012	Woehr et al.	N/A	N/A
8444605	12/2012	Kuracina et al.	N/A	N/A
8454536	12/2012	Raulerson et al.	N/A	N/A
8460247	12/2012	Woehr et al.	N/A	N/A
8469928	12/2012	Stout et al.	N/A	N/A
8496628	12/2012	Erskine	N/A	N/A
D687548	12/2012	Hayashi	N/A	N/A
8506533	12/2012	Carlyon et al.	N/A	N/A
8509340	12/2012	Michelitsch	N/A	N/A
8517959	12/2012	Kurosawa et al.	N/A	N/A
8529515	12/2012	Woehr et al.	N/A	N/A
8535271	12/2012	Fuchs et al.	N/A	N/A
8540728	12/2012	Woehr et al.	N/A	N/A
8545454	12/2012	Kuracina et al.	N/A	N/A
8568372	12/2012	Woehr et al.	N/A	N/A
8574203	12/2012	Stout et al.	N/A	N/A
8579881	12/2012	Agro et al.	N/A	N/A
8585651	12/2012	Asai	N/A	N/A
8585660	12/2012	Murphy	N/A	N/A
8591467	12/2012	Walker et al.	N/A	N/A
8591468	12/2012	Woehr et al.	N/A	N/A
8597249	12/2012	Woehr et al.	N/A	N/A
8622931	12/2013	Teague et al.	N/A	N/A
8622958	12/2013	Jones et al.	N/A	N/A
8622972	12/2013	Nystrom et al.	N/A	N/A
D700318	12/2013	Amoah et al.	N/A	N/A
8647301	12/2013	Bialecki et al.	N/A	N/A
8647313	12/2013	Woehr et al.	N/A	N/A
8647324	12/2013	DeLegge et al.	N/A	N/A
8652104	12/2013	Goral et al.	N/A	N/A
8657790	12/2013	Tal et al.	N/A	N/A
8672888	12/2013	Tal	N/A	N/A
8679063	12/2013	Stout	604/167.04	A61M 25/0097
8690833	12/2013	Belson	N/A	N/A
8715242	12/2013	Helm, Jr.	N/A	N/A
8721546	12/2013	Belson	N/A	N/A
8728030	12/2013	Woehr	N/A	N/A
8728035	12/2013	Warring et al.	N/A	N/A
8740859	12/2013	McKinnon et al.	N/A	N/A
8740964	12/2013	Hartley	N/A	N/A
8747387	12/2013	Belley et al.	N/A	N/A

8753317	12/2013	Osborne et al.	N/A	N/A
8764711	12/2013	Kuracina et al.	N/A	N/A
D710495	12/2013	Wu et al.	N/A	N/A
8814833	12/2013	Farrell et al.	N/A	N/A
D713957	12/2013	Woehr et al.	N/A	N/A
D714436	12/2013	Lee-Sepsick	N/A	N/A
8827965	12/2013	Woehr et al.	N/A	N/A
8845584	12/2013	Ferguson et al.	N/A	N/A
D715931	12/2013	Watanabe et al.	N/A	N/A
8864714	12/2013	Harding et al.	N/A	N/A
8900192	12/2013	Anderson et al.	N/A	N/A
8932257	12/2014	Woehr	N/A	N/A
8932258	12/2014	Blanchard et al.	N/A	N/A
8932259	12/2014	Stout	604/167.03	A61M 25/0693
8945011	12/2014	Sheldon et al.	N/A	N/A
8951230	12/2014	Tanabe	604/167.03	A61M 5/158
8956327	12/2014	Bierman et al.	N/A	N/A
8974426	12/2014	Corcoran et al.	N/A	N/A
8979802	12/2014	Woehr	N/A	N/A
8986227	12/2014	Belson	N/A	N/A
D726908	12/2014	Yu et al.	N/A	N/A
8998852	12/2014	Blanchard et al.	N/A	N/A
9005169	12/2014	Gravesen et al.	N/A	N/A
9011351	12/2014	Hoshinouchi	N/A	N/A
9011381	12/2014	Yamada et al.	N/A	N/A
D728781	12/2014	Pierson et al.	N/A	N/A
9022979	12/2014	Woehr	N/A	N/A
9033927	12/2014	Maan et al.	N/A	N/A
D733289	12/2014	Blanchard et al.	N/A	N/A
9044583	12/2014	Vaillancourt	N/A	N/A
D735321	12/2014	Blanchard	N/A	N/A
9089671	12/2014	Stout et al.	N/A	N/A
9089674	12/2014	Ginn et al.	N/A	N/A
9095683	12/2014	Hall et al.	N/A	N/A
9101746	12/2014	Stout et al.	N/A	N/A
9108021	12/2014	Hyer et al.	N/A	N/A
9114231	12/2014	Woehr et al.	N/A	N/A
9114241	12/2014	Stout et al.	N/A	N/A
9126012	12/2014	McKinnon et al.	N/A	N/A
9138252	12/2014	Bierman et al.	N/A	N/A
9138545	12/2014	Shaw et al.	N/A	N/A
9138559	12/2014	Odland et al.	N/A	N/A
RE45776	12/2014	Root et al.	N/A	N/A
D740410	12/2014	Korkuch et al.	N/A	N/A
9149625	12/2014	Woehr et al.	N/A	N/A
9149626	12/2014	Woehr et al.	N/A	N/A
9155863	12/2014	Isaacson et al.	N/A	N/A
9162036	12/2014	Caples et al.	N/A	N/A
9162037	12/2014	Belson et al.	N/A	N/A

9180275	12/2014	Helm	N/A	N/A
D746445	12/2014	Lazarus	N/A	N/A
9205231	12/2014	Call et al.	N/A	N/A
9216109	12/2014	Badawi et al.	N/A	N/A
9220531	12/2014	Datta et al.	N/A	N/A
9220871	12/2014	Thorne et al.	N/A	N/A
9220882	12/2014	Belley et al.	N/A	N/A
D748254	12/2015	Freigang et al.	N/A	N/A
9227038	12/2015	Woehr	N/A	N/A
9242071	12/2015	Morgan et al.	N/A	N/A
9242072	12/2015	Morgan et al.	N/A	N/A
RE45896	12/2015	Stout et al.	N/A	N/A
D748774	12/2015	Caron	N/A	N/A
D748777	12/2015	Uenishi et al.	N/A	N/A
D749214	12/2015	Uenishi et al.	N/A	N/A
D749727	12/2015	Wapler et al.	N/A	N/A
D751194	12/2015	Yu et al.	N/A	N/A
D752737	12/2015	Ohashi	N/A	N/A
9289237	12/2015	Woehr et al.	N/A	N/A
9308352	12/2015	Teoh et al.	N/A	N/A
9308354	12/2015	Farrell et al.	N/A	N/A
9320870	12/2015	Woehr	N/A	N/A
D755368	12/2015	Efinger et al.	N/A	N/A
9352119	12/2015	Burkholz et al.	N/A	N/A
9352127	12/2015	Yeh et al.	N/A	N/A
9352129	12/2015	Nardeo et al.	N/A	N/A
9358364	12/2015	Isaacson	N/A	A61M 39/0606
9370641	12/2015	Woehr et al.	N/A	N/A
9381324	12/2015	Fuchs et al.	N/A	N/A
9399116	12/2015	Goral et al.	N/A	N/A
9408569	12/2015	Andreae et al.	N/A	N/A
9421345	12/2015	Woehr et al.	N/A	N/A
9427549	12/2015	Woehr et al.	N/A	N/A
D775330	12/2015	Blennow et al.	N/A	N/A
9522254	12/2015	Belson	N/A	N/A
D776259	12/2016	Eldredge	N/A	N/A
9545495	12/2016	Goral et al.	N/A	N/A
9554817	12/2016	Goldfarb et al.	N/A	N/A
D779059	12/2016	Nino et al.	N/A	N/A
D779661	12/2016	McKnight et al.	N/A	N/A
9579486	12/2016	Burkholz et al.	N/A	N/A
9586027	12/2016	Tisci et al.	N/A	N/A
9592367	12/2016	Harding et al.	N/A	N/A
9616201	12/2016	Belson	N/A	N/A
9623210	12/2016	Woehr	N/A	N/A
9675784	12/2016	Belson	N/A	N/A
9687633	12/2016	Teoh	N/A	N/A
D791311	12/2016	Yantz	N/A	N/A
9707378	12/2016	Leinsing et al.	N/A	N/A

9717523	12/2016	Feng et al.	N/A	N/A
9717887	12/2016	Tan	N/A	N/A
9737252	12/2016	Teoh et al.	N/A	N/A
9750532	12/2016	Toomey et al.	N/A	N/A
9750928	12/2016	Burkholz	N/A	A61M 39/162
9757540	12/2016	Belson	N/A	N/A
9764085	12/2016	Teoh	N/A	A61M 39/0613
9764117	12/2016	Bierman et al.	N/A	N/A
9775972	12/2016	Christensen et al.	N/A	N/A
9782568	12/2016	Belson	N/A	N/A
9789279	12/2016	Burkholz et al.	N/A	N/A
9795766	12/2016	Teoh	N/A	N/A
9844646	12/2016	Knutsson	N/A	N/A
9861792	12/2017	Hall et al.	N/A	N/A
9872971	12/2017	Blanchard	N/A	N/A
D810282	12/2017	Ratjen	N/A	N/A
D815737	12/2017	Bergstrom et al.	N/A	N/A
9950139	12/2017	Blanchard et al.	N/A	N/A
9962525	12/2017	Woehr	N/A	N/A
10004878	12/2017	Ishida	N/A	N/A
10086171	12/2017	Belson	N/A	N/A
10232146	12/2018	Braithwaite et al.	N/A	N/A
10328239	12/2018	Belson	N/A	N/A
10357635	12/2018	Korkuch et al.	N/A	N/A
10384039	12/2018	Ribelin et al.	N/A	N/A
10426931	12/2018	Blanchard et al.	N/A	N/A
D870271	12/2018	Kheradpir et al.	N/A	N/A
D870883	12/2018	Harding et al.	N/A	N/A
10493262	12/2018	Tran et al.	N/A	N/A
10525236	12/2019	Belson	N/A	N/A
10688280	12/2019	Blanchard et al.	N/A	N/A
10688281	12/2019	Blanchard et al.	N/A	N/A
10722685	12/2019	Blanchard et al.	N/A	N/A
10806906	12/2019	Warring et al.	N/A	N/A
D914208	12/2020	Shabudin et al.	N/A	N/A
D917694	12/2020	Schneider et al.	N/A	N/A
D921884	12/2020	Tran et al.	N/A	N/A
D929580	12/2020	Bornhoft	N/A	N/A
D933216	12/2020	Gloess et al.	N/A	N/A
D933820	12/2020	Ota	N/A	N/A
D942621	12/2021	Cheng et al.	N/A	N/A
D944395	12/2021	Harris et al.	N/A	N/A
D950719	12/2021	Moore et al.	N/A	N/A
D952842	12/2021	Harris et al.	N/A	N/A
D954258	12/2021	Hang et al.	N/A	N/A
11389626	12/2021	Tran et al.	N/A	N/A
11400260	12/2021	Huang et al.	N/A	N/A
D964559	12/2021	Fujii et al.	N/A	N/A
D967408	12/2021	Tanaka et al.	N/A	N/A

D982741	12/2022	Lee-Sepsick et al.	N/A	N/A
D988509	12/2022	Ko	N/A	N/A
D1015525	12/2023	Fang	N/A	N/A
D1026213	12/2023	Healy et al.	N/A	N/A
D1037439	12/2023	Williams et al.	N/A	N/A
D1042801	12/2023	Sender et al.	N/A	N/A
D1042874	12/2023	Perera et al.	N/A	N/A
D1043969	12/2023	Howard-Sparks et al.	N/A	N/A
D1054556	12/2023	Bornhoft	N/A	N/A
D1069106	12/2024	Stats et al.	N/A	N/A
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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a division of U.S. patent application Ser. No. 16/696,844, filed Nov. 26, 2019, now U.S. Pat. No. 11,759,618, which is a division of U.S. patent application Ser. No. 15/702,537, filed Sep. 12, 2017, now U.S. Pat. No. 10,493,262, which claims the benefit of U.S. Provisional Patent Application No. 62/393,531, filed Sep. 12, 2016, and titled “BLOOD CONTROL FOR A CATHETER INSERTION DEVICE,” each of which is incorporated herein by reference in its entirety.

BRIEF SUMMARY
(1) Briefly summarized, embodiments of the present invention are directed to a tool for assisting with the placement into a patient of a catheter or other tubular medical device. In particular, a fluid control component configured for controlling fluid flow through the hub of the catheter assembly during and after placement into the patient is disclosed.
(2) In one embodiment, the fluid control component comprises a body disposed within a cavity of the hub, the body being movable between a first position and a second position, wherein the body does not pierce a valve disposed in the hub when in the first position and wherein the body pierces the valve when in the second position. The body includes a conduit to enable fluid flow through an internal portion of the body when the body is in the second position, and a plurality of longitudinally extending ribs disposed on an exterior surface of the body. The ribs provide at least

one fluid flow channel between the valve and an external portion of the body when the body is in the second position.
(3) These and other features of embodiments of the present invention will become more fully apparent from the following description and appended claims, or may be learned by the practice of embodiments of the invention as set forth hereinafter.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) A more particular description of the present disclosure will be rendered by reference to specific embodiments thereof that are illustrated in the appended drawings. It is appreciated that these drawings depict only typical embodiments of the invention and are therefore not to be considered limiting of its scope. Example embodiments of the invention will be described and explained with additional specificity and detail through the use of the accompanying drawings in which:
(2) FIGS. **1**A and **1**B are perspective views of a catheter insertion device according to one embodiment;
(3) FIG. **2** is an exploded view of the catheter insertion device of FIGS. **1**A and **1**B;
(4) FIG. **3** is a cross-sectional side view of a catheter according to one embodiment;
(5) FIG. **4** is a perspective view of a blood control component of the catheter of FIG. **3**;
(6) FIGS. **5**A and **5**B show various views of the blood control component of FIG. **4**;
(7) FIGS. **6**A-**6**C show various views of a blood control component for a catheter according to one embodiment;
(8) FIGS. **7**A-**7**C show various views of use of the catheter and blood control catheter of FIGS. **6**A-**6**C;
(9) FIGS. **8**A-**8**F show various views of a needle hub of the catheter insertion device of FIGS. **1**A and **1**B;
(10) FIG. **9** is a cross sectional view of a flash indicator of the catheter insertion device of FIGS. **1**A and **1**B;
(11) FIG. **10** is a perspective view of a valve of the catheter insertion device of FIGS. **1**A and **1**B; and
(12) FIG. **11** is an isolation view showing the blood control component piercing the valve of the catheter insertion device of FIGS. **1**A and **1**B.
DETAILED DESCRIPTION OF SELECTED EMBODIMENTS
(13) Reference will now be made to figures wherein like structures will be provided with like reference designations. It is understood that the drawings are diagrammatic and schematic representations of exemplary embodiments of the present invention, and are neither limiting nor necessarily drawn to scale.
(14) For clarity it is to be understood that the word “proximal” refers to a direction relatively closer to a clinician using the device to be described herein, while the word “distal” refers to a direction relatively further from the clinician. For example, the end of a catheter placed within the body of a patient is considered a distal end of the catheter, while the catheter end remaining outside the body is a proximal end of the catheter. Also, the words “including,” “has,” and “having,” as used herein, including the claims, shall have the same meaning as the word “comprising.”
(15) Embodiments of the present invention are generally directed to a tool for assisting with the placement into a patient of a catheter or other tubular medical device. For example, catheters of various lengths are typically placed into a body of a patient so as to establish access to the patient's vasculature and enable the infusion of medicaments or aspiration of body fluids. The catheter insertion tool to be described herein facilitates such catheter placement. Note that, while the discussion below focuses on the placement of catheters of a particular type and relatively short

length, catheters of a variety of types, sizes, and lengths can be inserted via the present device, including peripheral IVs, intermediate or extended-dwell catheters, PICCs, central venous catheters, etc. In one embodiment, catheters having a length between about 1.25 inch and about 2.25 inches can be placed, though many other lengths are also possible.

(16) FIGS. **1**A-**2** depict various details regarding a catheter insertion tool ("insertion tool" or "insertion device"), generally depicted at **10**, according to one embodiment. As shown, the insertion tool **10** includes a housing **12** that may itself include a proximal housing portion **12**A and a distal housing portion **12**B. The housing **12** further includes an open distal end, and can include a flat bottom to enable the insertion device **10** to lie flat on a surface without tipping. In another embodiment, the housing is integrally formed. In yet another embodiment, a top housing portion and a bottom housing portion can be employed, or more than two portions can be used. In the present embodiment, the housing composed of a thermoplastic such as polycarbonate and is translucent, though other configurations are contemplated. The housing **12** defines grip surfaces **13** on either side of the housing, as seen in FIGS. **1**A and **1**B, to enable grasping of the insertion device **10** by the user.

(17) A needle hub **14** supporting a hollow needle **16** (which together form part of a needle assembly, in one embodiment) is included with the housing **12**. In the present embodiment, the needle hub **14** is secured within the housing **12** within a cavity **13** defined by the housing, but in another embodiment it can be integrally formed with the housing.

(18) As will be described in detail further below, the needle hub **14** includes a slot for receiving a portion of the needle **16** and a quantity of adhesive, such as liquid or UV-cure adhesive for example, in order to fix the needle in place in the needle hub. The needle **16** extends distally from the needle hub **14** so as to extend past the distal end of the distal housing portion **12**B and terminates at a distal end **16**B thereof. A notch **18** is defined through the wall of the needle **16** proximate the distal end thereof. The notch **18** enables flashback of blood to exit the lumen defined by the hollow needle **16** once access to the patient's vasculature is achieved during catheter insertion procedures. Thus, blood exiting the notch **18** can be viewed by a clinician to confirm proper needle placement in the vasculature, as will be explained further below.

(19) A catheter **42** including a catheter tube **44** is removably disposed on the portion of the needle **16** residing external to the housing **12** such that the needle occupies a lumen of the catheter defined by a catheter tube. The catheter tube **44** extends distally from a hub **46** of the catheter **42**, which hub is initially disposed adjacent the open distal end of the distal housing portion **12**B, as shown in FIGS. **1**A and **1**B.

(20) The insertion device **10** further includes a guidewire advancement assembly **20** for advancing a guidewire **22** through the needle **16** and into the vasculature of the patient once vessel access by the needle has been achieved. The guidewire **22** (FIGS. **1**A-**2**) is pre-disposed within the lumen of the needle **16**. The guidewire advancement assembly **20** includes a guidewire lever **24** that selectively advances the guidewire **22** in a distal direction during use of the insertion device **10** such that the distal portion of the guidewire extends beyond the distal end **16**B of the needle **16**. In the present embodiment, a finger pad **28** of the guidewire lever **24** is slidably disposed on the housing **12** via a slot **32** to enable a thumb and/or finger(s) of the user to selectively advance the guidewire **22** distally past the distal end **16**B of the needle **16**. Of course, other engagement schemes to translate user input to guidewire movement could also be employed. In one embodiment, the guidewire **22** can include a guidewire support tube to provide additional stiffness to the guidewire and facilitate its distal advancement described above. In yet another embodiment, a proximal end of the guidewire can be attached at an anchor point on an interior portion of the housing **12** (or other fixed portion of the insertion device **10**) and looped about a proximal portion of the guidewire lever **24** in a roughly U-shaped configuration such that the distal end of the guidewire extends two units of distance distally past the distal end **16**B of the needle **16** for every one unit of distance of movement of the finger pad **28**. These and other modifications are therefore

contemplated.
(21) The majority length of the guidewire in one embodiment includes a metal alloy of nickel and titanium commonly referred to as nitinol, though other suitable guidewire materials can also be employed.
(22) FIGS. **1**A and **1**B show that the catheter **42** is removably attached to the insertion device **10** such that the catheter tube **44** thereof is disposed over the portion of the needle **16** that extends distal to the housing **12** such that the catheter resides external to the insertion device housing. The catheter **42** in the present embodiment is kept in place against the open distal end of the housing via a friction fit with one or more features disposed on the housing distal end. A tab **48** is included on the catheter hub **46** for assisting with manual distal extension of the catheter **42** by a user during deployment thereof.
(23) Note that in one embodiment the outer diameters (and/or other areas) of the needle **16** and the catheter tube **44** are lubricated with silicone or other suitable lubricant to enhance sliding of the catheter tube with respect to the needle and for aiding in the insertion of the catheter into the body of the patient.
(24) The insertion device **10** includes a retraction system configured to selectively retract the needle **16** into the housing **12**. In detail, a spring element, such as a coil spring **50**, is disposed between a distal end of the inner cavity **13** of the housing **12** and a ridge **144** disposed at a proximal end of the needle hub **14**. The spring **50** is disposed about the needle hub **14**, and the needle hub is proximally slidable within the cavity of the housing **12**. The needle hub **14** is kept in a distal position within the cavity of the housing **12**, with the spring maintained in a compressed configuration, by a retraction button **52** disposed near the distal end of the housing **12**. Manual depression of the retraction button **52** releases engagement of the retraction button with the needle hub **14**, which in turn enables the spring **50** to expand, causing the needle hub to move proximally within the cavity of the housing **12**. This in turn retracts the needle **16** so that the distal end **16**B thereof is retracted into the housing **12** and protected from inadvertent contact by a user. Note that other needle safety configurations can also be employed.
(25) Use of the insertion device **10** in placing the catheter **42** in the vasculature of a patient is described here. A user grasping the insertion device **10** first guides the distal portion of the needle **16** through the skin at a suitable insertion site and accesses a subcutaneous vessel. After needle access to the vessel is confirmed, the guidewire advancement assembly **20** is actuated, wherein the finger pad **28** (disposed in the slot **32** defined in the housing **12**) is advanced by the finger of the user to distally advance the guidewire **22** (FIG. **3**), initially disposed within the hollow needle **16**. Note that the guidewire **22** is distally advanced by the guidewire lever **24**, which is operably attached to the slidable finger pad **28**.
(26) Distal advancement of the guidewire **22** continues until the finger pad **28** has been distally slid a predetermined distance, resulting in a predetermined length of the guidewire **22** extending past the distal end of the needle **16**, as shown in FIGS. **1**A and **1**B. This places the distal portion of the guidewire **22** within the vessel.
(27) Once the guidewire lever **24** has been fully distally extended via sliding of the finger pad **28**, which in turn has extended the guidewire **22** past the distal end **16**B of the needle **16**, manual distal advancement of the catheter **42** is performed, using the tab **48** of the catheter hub **46**, which causes the catheter tube **44** to slide over distal portions of the needle **16** and guidewire **22** and into the patient's vasculature via the insertion site. In light of this, it is appreciated that the finger pad **28** acts as a first member used to advance the guidewire **22**, whereas manual advancement is employed to advance the catheter **42**, in the present embodiment. In another embodiment, it is appreciated that the finger pad **28** can be employed to also distally deploy the catheter **42** at least a partial distance into the vessel.
(28) The catheter **42** is distally advanced until it is suitably disposed within the vessel of the patient. The retraction button **52** on the housing **12** is then manually depressed by the user, which

causes the spring **50** to decompress and retract the needle hub **14**, which in turn causes the distal end **16**B of the needle **16** to be retracted within the housing **12** and preventing its re-emergence, thus protecting the user from accidental needle sticks. Thus, this serves as one example of a needle safety component, according to the present embodiment; others are possible. The catheter **42** is physically separated from the housing **12** at this time. Now in place within the patient, the catheter **42** can be prepared for use and dressed, per standard procedures. Then insertion device **10** can be discarded.

(29) In additional detail, FIG. **2** shows a continuous blood flash indicator **80** that can be used with the insertion device **10** according to one embodiment. The flash indicator **80** is employed to indicate the presence of blood in the lumen of the needle **16** during use of the device **10**, thus assuring that proper access has been made by the needle into a vein or other desired blood-carrying vessel. As shown in FIG. **9**, the flash indicator **80** includes a translucent chamber **82** that is generally cylindrical in shape, sealed at either end, and disposed about a portion of the needle **16** such that the needle protrudes out from either sealed end. In the present embodiment the chamber **82** of the flash indicator **80** is disposed in the slot **142** (FIGS. **8**A-**8**F) of the needle hub **14** within the housing **12**, though other locations along the needle are also possible.

(30) Two notches—a first notch **83** and a second notch **84**—are defined in the needle **16** so as to provide fluid communication between the lumen of the needle and the interior of the flash indicator chamber. The notches **83**, **84** replace the notch **18** (FIG. **2**) in one embodiment, and are included in addition to the notch **18**, in another embodiment. It is appreciated that, in one embodiment, blood passage through the notch **18** serves as an initial indicator that the distal end **16**B of the needle has entered the vein, while the embodiment shown here serves as an additional indicator to verify that the needle distal end remains in the vein after initial access. Further detail regarding the flash indicator 80 can be found in U.S. Publication No. 2016/0331938, published Nov. 17, 2016, and entitled “Catheter Placement Device Including an Extensible Needle Safety Component,” which is incorporated herein by reference in its entirety.

(31) In the present embodiment, the guidewire **22** passes through the lumen of the needle **16** so as to extend through the flash indicator **80**. The first notch **83** is disposed distal to the second notch **84** toward the distal end of the chamber **82**.

(32) When vessel access is achieved by the distal end **16**B of the needle **16**, blood travels proximally up the lumen of the needle, between the inner surface of the needle and the outer surface of the guidewire **22**, disposed in the needle lumen (FIG. **9**). Upon reaching the relatively more distal first notch **83** defined in the needle **16**, a portion of the blood will pass through the first notch and enter the chamber **82**. As the blood fills the translucent chamber **82**, a user can observe the chamber through the translucent housing **12** of the insertion device and view the blood therein, thus confirming that the vessel access has been achieved. In another embodiment, the housing **12** can be configured such that direct viewing of the chamber **82** is possible, e.g., with no intervening structure interposed between the chamber and the user.

(33) The second notch **84** is employed to provide an exit point for air in the chamber **82** to equalize air pressure and enable the blood to continue entering the chamber via the first notch **83**. It is noted that the spacing between the inner surface of the needle **16** and the outer surface of the guidewire **22** along section **85** is such that air but not blood can pass therebetween, thus enabling air pressure equalization in the chamber without blood passage through the second notch **84**. In this way, the flash indicator **80** is a continuous indicator, enabling a continuous flow of blood into the chamber **82** while the needle distal end **16**B is disposed within the blood-carrying vessel.

(34) Note that the catheter insertion device **10** can include more than one flash indicator. In one embodiment and as mentioned above, for instance, the blood flash indicator **80** can be included, along with another flash indicator, such as the notch **18**, which enables blood present in the lumen of the needle **16** to proceed proximally up the space between the outer surface of the needle and the inner surface of the catheter **42**.

(35) FIG. **3** depicts various details of a blood control component **100** included with the catheter **42**, in accordance with one embodiment. As shown, the blood component **100** is slidably disposed within a cavity **46**A of the catheter hub **46** and is configured to selectively enable fluid flow through the catheter **42** in concert with a valve **102**, also disposed within the catheter hub cavity. The valve **102** in the present embodiment is a tricuspid valve including three leaflets **102**A defined by a plurality of slits **103** as seen in FIG. **10**, though other valve types may also be employed.
(36) FIGS. **4-5**B depict various details of the blood control component **100**, including an elongate body **104** extending between a proximal end **104**A and a distal end **104**B and defining a central conduit **106** through which fluids can flow. A plurality of ribs **110** is disposed on an outer surface of the body **104** such that the ribs longitudinally extend from proximally past the proximal end **104**A of the body to the distal end **104**B thereof. Each rib **110** radially extends from the body **104** to define a contoured profile along the longitudinal length thereof. The body **104** and ribs **110** contribute to generally define a conical shape to the blood control component **100**. Deviations from the conical shape are also possible in other embodiments.
(37) Each rib **110** further defines a notch **112** intermediately positioned along the longitudinal length of the rib, as well as a protrusion **114** at the proximal end of the rib. As seen in FIG. **3**, the notch **112** of each rib **110** receives a portion of an annular ridge **120** defined on an inner surface of the catheter hub cavity **46**A to keep the blood control component **100** in place within the cavity before actuation. Correspondingly, the protrusions **114** of each rib **110** engage with the annular ridge **120** when the blood control component **100** is actuated so as to prevent further distal movement thereof past its intended length of travel. The body **104** defines a channel **126** between adjacent ribs **110**, thus providing four fluid flow channels in the illustrated embodiment. Note that in one embodiment one or more ribs **110** can be offset along the longitudinal length thereof such that a proximal portion of the rib including the protrusion **114** is not longitudinally aligned with (as in FIGS. **5**A and **5**B), but rather circumferentially offset from, a more distal portion of the rib.
(38) FIGS. **6**A-**6**C depict details of the blood control component **100** according to another embodiment, wherein the body **104** defines a plurality of channels **126** disposed about the conduit **106**. An intermediate, annular shoulder **128** is also defined by the body **104**.
(39) FIGS. **7**A-**7**C depict various stages of operation of the blood control component **100** of FIGS. **6**A-**6**C, though the principles described here also apply to the embodiment shown in FIGS. **4-5**B as well. In particular, FIGS. **7**A and **7**B show the blood control component **100** in a relatively proximal position, also referred to herein as an un-actuated state, wherein the annular ridge **120** is received within the notches **112** (below the shoulders **128**) of each rib **110** of the blood control component. In this position, the distal end **104**B of the body **104** does not protrude through the valve **102** that is positioned distal to the blood control component and thus no fluid is able to pass through the catheter **42**, as desired. The valve **102** in the closed position thus prevents blood leakage through the catheter **42**, such as when the catheter has been placed within the patient but no external connection has been made to the catheter hub **46**.
(40) In contrast, FIG. **7**C shows the blood control component **100** in a relatively distal position, also referred to herein as an actuated state, wherein the blood control component has been distally advanced (such as by insertion of a male luer connector into the catheter hub **46**) such that the distal end **104**B thereof has penetrated through the leaflets **102**A of the valve **102**, thus providing a fluid path through the valve via the conduit **106** of the blood control component. Further distal advancement of the blood control component **100** is prevented by engagement of the protrusions **114** against the annular ridge **120**. As mentioned, the distal movement of the blood control component **100** is caused by the insertion into the catheter hub cavity **46**A by a luer connector or other apparatus that can be operably connected to the catheter hub **46**.
(41) In accordance with the present embodiment, the blood control component **100** is configured to eliminate an entrapment zone between the blood control component and the valve **102** after the blood control component has pierced the valve in its actuated state. Specifically, and with respect to

the embodiment shown in FIGS. **4-5**B, the ribs **110** cause additional deformation of the leaflets **102**A of the valve **102** when the blood control component pierces the valve, as seen in FIG. **11**. This in turn prevents partial sealing of the leaflets **102**A to the exterior surface of the blood control component body **104**, thus providing spacing therebetween and additional fluid flow paths via the channels **126** between the exterior surface of the blood control component body **104** and the valve leaflets. Thus, fluid is able to flow through the catheter hub cavity **46**A not only internal to the blood control component body **104** via the conduit **106** but also external to the blood control component body via the channels **126**, which are made patent by the interaction of the ribs **110** with the valve leaflets **104**A. This fluid flow external to the blood control component **100** assists in moving fluid through the entirety of the hub cavity **46**A, thus desirably preventing fluid flow stagnation in the region between the blood control component **100** and the valve **102**.

(42) Note that in the present embodiment an outer termination point of each slit **103** that form the leaflets **102**A defines a staggered termination point, as seen in FIG. **11**. Note also that the ribs described herein are but one example of one or more extended surfaces that can be included with the blood control component to enable additional fluid flow channels to be defined on an outer surface of the blood control component to enable fluid flow about the exterior of the blood control component when the blood control component pierces the valve. Examples of other extended surfaces include bumps, annular surfaces, fins, etc. These and other embodiments are therefore contemplated.

(43) The blood control component **100** of FIGS. **6**A-**6**C operates similarly to that described immediately above in connection with FIGS. **4-5**B, wherein the channels **126** provide fluid flow in addition to the conduit **106** so as to prevent fluid flow stagnation between the blood control component **100** and the valve **102**.

(44) FIGS. **8**A-**8**F depict various details regarding the aforementioned needle hub **14** of the insertion device **10**, which includes an elongate body **140** extending between a proximal end **140**A and a distal end **140**B. A slot **142** extends longitudinally along the length of the body **140** and is sized for receiving a portion of the length of the needle **16** therein. As mentioned, the ridge **144** is included on the proximal end **140**A of the needle hub and provides a surface against which the spring **50** can act to retract the needle hub and attached needle **16** into the cavity of the housing **12**. The slot **142** defines a volume **146** within which the above-described flash indicator **80** can be received.

(45) Note that the slot **142** is configured so that differing sizes of needle can be received and affixed therein. To that end, the slot **142** includes three shoulders **148** to support the needle **16** within the slot **142**. Note that the proximal edge of each of the shoulders **148** is relatively abrupt in shape so as to prevent spillage of a liquid epoxy adhesive that is placed in the slot **142** proximate the shoulders to secure the needle **16** within the slot.

(46) Embodiments of the invention may be embodied in other specific forms without departing from the spirit of the present disclosure. The described embodiments are to be considered in all respects only as illustrative, not restrictive. The scope of the embodiments is, therefore, indicated by the appended claims rather than by the foregoing description. All changes that come within the meaning and range of equivalency of the claims are to be embraced within their scope.

Claims

1. An insertion device, comprising: a housing, comprising: a needle including a distal notch in a sidewall of the needle, the distal notch proximate a distal end of the needle, and a pair of proximal notches in the sidewall of the needle proximate a proximal end of the needle, wherein the distal notch and the pair of proximal notches are in fluid communication with a lumen of the needle; a flash chamber enclosing the pair of proximal notches; and a needle hub including a slot for receiving a proximal portion of the needle and the flash chamber; and a catheter assembly,

comprising: a catheter including a catheter hub coupled to a proximal end of a catheter tube, the catheter hub including a valve and an annular ridge defined on an inner surface of the catheter hub proximal of the valve; and a fluid control component disposed in the catheter hub, the fluid control component movable between a first position and a second position, the fluid control component including a plurality of longitudinal ribs spaced apart on an exterior surface of the fluid control component, each of the plurality of longitudinal ribs extending from a proximal end of the fluid control component to a distal end of the fluid control component, each of the plurality of longitudinal ribs comprising: a notch at an intermediate portion between the proximal end and the distal end, the notch configured to engage the annular ridge in the first position; and a protrusion at the proximal end configured to engage the annular ridge in the second position.
2. The insertion device according to claim 1, wherein the valve of the catheter includes a tricuspid valve including three slits.
3. The insertion device according to claim 2, wherein an outer termination point of each slit is a staggered termination point.
4. The insertion device according to claim 1, wherein each of the plurality of longitudinal ribs includes a shoulder adjacent to the notch.
5. The insertion device according to claim 1, wherein the fluid control component has a generally conical shape.
6. The insertion device according to claim 1, wherein the plurality of longitudinal ribs comprise four equally spaced longitudinal ribs.
7. The insertion device according to claim 1, wherein each of the plurality of longitudinal ribs extend proximal of a proximal opening of the fluid control component.
8. The insertion device according to claim 1, wherein the fluid control component further comprises a fluid flow channel between each of the plurality of longitudinal ribs.
9. The insertion device according to claim 1, wherein the flash chamber is formed from a translucent material.
10. The insertion device according to claim 1, wherein the slot of the needle hub includes a plurality of shoulders configured to support the needle, wherein a proximal shoulder includes a proximal edge configured to prevent a flow of a liquid adhesive out of the slot when the liquid adhesive is used to secure the needle in the slot.
11. The insertion device according to claim 1, wherein the needle hub includes a radially extending ridge disposed proximate a proximal end of the needle hub, the radially extending ridge configured to support a spring of the insertion device.
12. The insertion device according to claim 11, further comprising a retraction system configured to selectively retract the needle into the housing, the retraction system comprising the spring and a retraction button in releasable engagement with the needle hub.
13. The insertion device according to claim 1, further comprising a guidewire advancement assembly including a guidewire pre-disposed in the lumen of the needle.
14. The insertion device according to claim 13, wherein the guidewire advancement assembly further includes a finger pad coupled to the guidewire, wherein the guidewire is designed for distal advancement upon distal advancement of the finger pad.